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Enhancing architectural education through hybrid digital tools: investigating the impact on design creativity and cognitive processes

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Abstract

Architectural education has been transformed by digital tools, yet significant gaps remain in their application, particularly in supporting the iterative and creative aspects of design thinking. This study aims to investigate the impact of hybrid digital tools on enhancing architectural education and addressing these gaps in integrated design studio at an accredited program. A multi-method qualitative approach was employed to collect research data, including semi-structured interviews with 23 architects and academic researchers and surveys of 50 architecture students. Thematic analysis was used to analyze the data. The findings revealed that while digital tools positively influence the design thinking process and creativity, limitations in interaction methods hinder the ability to reach comprehensive design solutions. Integrating hybrid digital toolkits was found to enhance designer-tool interaction, enabling the seamless inclusion of design thinking stages in an integrated manner and positively impacting creativity across various design stages. This study uniquely contributes by proposing strategies to integrate hybrid toolkits into architectural curricula, bridging the gap between traditional and digital methods and better preparing students to tackle complex design challenges using advanced methodologies. These findings highlight the need for educational frameworks that incorporate hybrid digital tools, stimulate innovation and better prepare students to tackle complex design challenges using advanced technologies.

Keywords: Architectural education, Hybrid digital tools, Integrated model, Design cognition, Response

Introduction

Technological advancements significantly affect the field of architecture, expanding the architect's toolkit and transforming the design process. In recent years, the emphasis on digital software-assisted architectural design has shifted from being viewed as a "mere technical problem" confronting architectural design to representing a sociocultural shift in design (Yin et al., 2015) This shift has led to the emergence of new design methodologies, processes, and strategies that aim to link design methods to the growth of digital

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design theory (Garg et al., 2022) However, there is still a misunderstanding about the impact of these digital software programs on design practice and education (Yang, 2000).

In the context of implementing new technologies to create architectural designs and integrating them throughout design processes, understanding how people interact with digital tools provides an adequate foundation for acquiring knowledge in the design disciplines. Research in this area has shown that digital tool systems have a beneficial impact on the creative work of designers (Oxman, 2006). However, these tools are currently seen as insufficient for aiding in the early stages of the design process, where the conclusions of many studies draw attention to the paradoxical nature of the digital tool, as they aim to enhance and expand traditional design practices but can unintentionally have unintended impediments and constraints on design processes, especially with the obstacles and shortcomings facing current architectural education (Cross, 2021; Pugh, 1991; Yang, 2000).

Several shortcomings impede the success of digital tools to support early design stages (Cross, 2021; Jones, 1992; Pugh, 1991; Zboinska, 2019). One issue is that digital tools focus on digitizing conceptual design, whereas designers often prefer to work with traditional tools at those stages. Even though digitization can be beneficial early on, practically for the time-consuming processes of human designers where manual drawing tools fail to formulate complex design ideas, current digital tool systems require intuitive input or output devices for conceptual design, limiting designers interaction with the design medium and restricting human creativity (Cross, 2021; Zboinska, 2019). Information synchronization and communication during conceptualization typically require more flexible and intuitive digital design tools (Cross, 2021; Jones, 1992; Pugh, 1991; Zboinska, 2019).

Another possible explanation is that digital systems do not perform well in initial design because they struggle to handle the complexities of human design thinking. The design process is non-linear, iterative, and collaborative. However, most present digital design systems provide strategies based on linear or downward design models. Even though repeating is feasible, it is often a computer-driven process with little room for human intervention. One system cannot support all of the cognitive processes of ideation. The early design process is divided into several stages. Digital tools generally focus on one of these stages, which is frequently "detailed design." Therefore, digital tools serve as tools to represent final designs rather than as facilitators aimed at activating creativity (Design as a Discipline, 2010; Liu et al., 2003; Oxman, 2006; Yang, 2000; Yin et al., 2015).

In The context of architectural education, there are five major issues: a lack of collaboration between acquired knowledge and applied design studios, students' familiarity with the effects of design elements on users, and students' failure to think about and realize their design process. a lack of guiding experimental learning, and insufficient opportunities and ways to acquire occasional knowledge and experience in designing problems and solutions (Cross, 2010; Sarkar & Chakrabarti, 2007; Yu et al., 2022). Recent research suggests that integrating distinct digital systems into hybrid new entities can mitigate these shortcomings. Integration of computer-aided design (CAD) systems with other systems, such as computer-aided engineering (CAE) systems, computer-aided manufacturing (CAM) systems, and extended reality (XR) systems has a positive impact on the creative behaviors of designers (Soliman et al., 2019; Yu et al., 2021) Many studies in this field

show that these integrations positively impact designers' creative behaviors. However, little research has been conducted on this type of incorporation (Grobman et al., 2010; Liu et al., 2003) while Oxman (2006) presents a theoretical model for such a system, he only provides a high-level overview of the integration. Thus, research is needed into how this fusion affects the human-tool relationship and designers' creative behaviors. Based on many studies, the correct understanding of teaching methods in architecture schools can be achieved by analyzing the current situation and determining the level of overlap between application and digital design tools through theoretical and analytical investigation. This understanding is a starting point for discovering when, where, and how digital design tools and applications should be integrated into architectural education.

Problem statement

Architectural education currently faces challenges in integrating digital tools effectively during the early design stages. These tools often prioritize detailed design stages, neglecting conceptual design's iterative and creative activities. These tools also limit the designers' thinking and creative processes, leading them to switch between digital and traditional tools during this critical phase, limiting their ability to fully engage with the design process. Additionally, there is a lack of empirical studies that address the integration of hybrid digital toolkits within architectural education. This study aims to bridge this gap by investigating how hybrid digital tools influence design creativity and cognitive processes in architectural education.

Research statement and objective

This research aims to investigate the impact of using hybrid digital tools on design creativity and cognitive processes in architectural education. In order to achieve this, the study has the following objectives: Evaluate the impact of hybrid digital tools on design creativity and cognitive processes in architectural education; Explore strategies for the integration of these tools in architectural curricula; and develop guidelines to optimize the use of the tools to enhance creativity, improve student outcomes, and address current limitations in tool utilization.

Research questions

How do hybrid digital tools affect the design thinking process in architectural education?

What are the limitations of current digital tools in fostering creativity, and how can hybrid tools overcome these challenges?

How can the findings inform the development of a more effective educational framework for integrating hybrid digital tools into architectural curricula?

Literature review

This section presents the theoretical foundations of the research, which also identifies the creative behaviors of the architects that must be considered, and presents the theoretical foundation for designing the tool to assess the mechanism and impact of using the Hybrid Digital Toolkit, describing its components and providing justifications for their integration.

Behaviors to enhance the design creativity

To investigate whether and how the hybrid digital design toolset affects the behaviors of architects, it is essential to define the relevant behaviors. This study focuses on the activities that might be able to make up for the drawbacks of the digital design assistance technologies listed in the introduction. It studied two types of behaviors: interactions between the designer and digital tool and design thinking.

Interactions between the designer and digital tool

Yin et al. (2015) pointed out that digital design tools should be characterized by integration between humans and tools, where the designer can cooperate smoothly with the machine to compensate for the shortcomings of both humans and tools. In this way, digital applications perform the quantitative tasks, and the designer participates in the qualitative activities, enhancing human creativity and alleviating the problems associated with digital applications (Yin et al., 2015). Studies indicated that human–machine integration should take place in three forms to ensure coherent and stronger integration: human-dominated, tool-dominated, and balanced human-tool cooperation. This approach allows for different types of interaction during the different design stages (Cross, 2021; Yang, 2000). An effective digital design system must enable the designer to freely enter multiple interactions with digital tools and utilize human perception and computing interchangeably (Eren & Yılmaz, 2022; Zboinska, 2019).

Design thinking activities and digital design thinking

Previous studies revealed that creative design includes three types of design thinking activities and stages, leading to the gradual generation of creative design concepts: divergence, transformation, and convergence. Where divergence involves the creation of multiple design solutions, transformation refers to the activity of optimizing design solutions, and convergence involves choosing the final design (Creely & Henriksen, 2019; Jones, 1992; Pugh, 1991) Studies indicated that in the context of creative design using digital design tools, these stages must be implemented in several iterative cycles to ensure an appropriate, meaningful, and viable final design concept (Grobman et al., 2010; Liu et al., 2003; Xiang et al., 2020a; Zboinska, 2019) Therefore, successful digital design systems must enhance designers' ability to generate ideas at all design development levels to improve designers' creative thinking behaviors and allow exploration and improvement of design solution possibilities at all levels of idea abstraction (Sarkar & Chakrabarti, 2007).

Digital design methods and tools have become essential in the design process, leading to new design processes and redefinition of design thinking. This new paradigm, termed digital design thinking, goes beyond representation and purposeful manipulation directed towards symbolic representations, creating a new vision of design thinking that is non-deterministic and non-stereotypical. More than a formal set of preferences or rejection of conventional forms of formal, stereotyped knowledge. Digital design thinking explores new forms and relationships between the designer, the image, and the information. According to Oxman (2006), new relationships are needed to find new forms

and vocabulary and create new approaches and ways to deal with design. Understanding and assimilating complexity is one of the most important features of digital design; integration provides new ways of thinking about design.

Digital design model

As reliance on digital design increases to develop the effectiveness and efficiency of architectural products, design, and production practices in general in education and professionalism, several models have emerged to interpret design using digital tools. Oxman (2006) provided a diagram of five digital design models: the two-process CAD model, shaping–formation models, Generative models, performance models (including performance-based configuration and generation), and the integrated compound model. These models illustrate the development of digital tools and applications, their potential as integrated and interactive design tools, and their impact on design processes from the visualization to the construction process.

Based on the studies, composite models are considered the future goal of all digital systems; they integrate design processes, including formation, generation, evaluation, and performance, to provide a flexible flow of data and information in multiple directions during various design stages (Cross, 2010; Lawson, 2004; Oxman, 2006) this complex integrated network enhances and increases the performance of design media.

The computational design model was introduced by (Yu et al., 2021), which includes complex relationships between design cognition, design environment, design technology, and design creativity. The model defines the design environment as a set of conditions that affect the mechanism of the designer's work and interaction with the design. Design perception is described as behaviors and intangible mental processes during the design process. Design technology is defined as the tools that allow the designer to visualize, create, and analyze the various design elements, including developments in technical tools, devices, and interfaces, in addition to the factors that contribute to the perception of external design (Yu et al., 2021, 2022; Zboinska, 2019) The model distinguishes between design technology, which supports and enhances design processes, and design perception, which refers to the natural processes occurring during the design process, Highlighting the design creation process as the central goal, the model emphasizes the importance of integrating these elements to achieve comprehensive design solutions (Yu et al., 2022).

Nature of the design problem

Cross (2010) and Lawson (2006) discussed that the design problem is characterized as "ill-defined" with various constraints existing in a multidimensional form and, therefore, cannot be addressed comprehensively. Therefore, designers must use their knowledge and experience to understand and develop problem characteristics, giving structure to unstructured problems and reformulating them to promote the development of solutions (Cross, 2021; Demir et al., 2016; Lawson, 2004).

Lawson's (2004) model integrates formulation, movement, representation, evaluation, and reflection with the cognitive processes essential for developing comprehensive design solutions. Integrating these stages with design constraints is necessary to achieve successful design solutions (Lawson, 2004, 2006) Cross (2011) showed that the desired

design solution is a comprehensive response involving investigating problems from the perspective of both small details and large issues, integrating discovered limitations to create integrated and comprehensive solutions.

Hybrid digital toolkit

Oxman's composite integrated digital design model assumes that the designer interacts with five different 3D modeling methods: shaping, animation, specific design, generative modeling, and performance design. This model supports designers' diverse creation needs at different stages of design while maintaining control over design processes (Oxman, 2006) The model allows designers to direct interactions with digital tools, whether human-dominated, tool-dominated, or both. This flexibility increases creativity opportunities in the design process(Soliman et al., 2019) Zboinska (2015) investigated the idea of applying a diverse and extensive digital toolkit to enhance the architectural design processes (Soliman et al., 2019; Yu et al., 2022; Zboinska, 2015).

Design creativity research suggests that the large design solution space formed by the hybrid toolkit and its wide range of design possibilities may result in superior creative and innovative designs (Eren & Yılmaz, 2022). Additionally, using various digital tools has some intriguing ramifications for how they perceive design. Using a wide variety of digital tools significantly alters the function of tools. When used in conjunction with design thinking processes, the tools turn into genuine collaborators of the designer (Xiang et al., 2020a). The integration of comprehensive digital teaching has become the main topic of discussion among instructors and researchers nowadays. Digital tools improve student's skills and abilities and enhance academic performance and output. However, many colleges of architecture, globally and locally, have attempted to integrate modern digital design tools into experimental curricula. However, the lack of a basic framework for correctly using digital design tools remains a challenge. The lack of integration of technology and design features in architectural education as a mixture that students deal with in a real setting, in addition to the lack of proper teaching guidance, continues to be a significant issue (Design as a Discipline, 2010; Garg et al., 2022; Neroni et al., 2021; Xiang et al., 2020a).

Existing research highlights several shortcomings of digital tools in supporting early-stage of design (Cross, 2021; Oxman, 2006). The most significant shortcomings relate to how digital tools are integrated into architectural education and practice, particularly in fostering creativity and adapting to evolving industry needs. Existing digital tools, while advancing in technical capabilities, often struggle to support the iterative and nonlinear nature of early-stage design processes. This limits the potential for fostering creativity, as many tools emphasize automation over intuitive, hands-on interactions. Furthermore, architectural education has not fully addressed the pedagogical strategies required to teach digital tools like BIM and computational design methods effectively. While students demonstrate positive attitudes toward these tools, there is a lack of systematic research on their impact on learning outcomes and creative skill development (Kharvari & Kaiser, 2022; Q. Yang et al., 2023).

Furthermore, these tools have limited capabilities to align with sustainable design practices. Although previous studies investigate the technical functionalities of digital tools such as building information modeling (BIM) and building performance analysis

(BPA), they fail to address architects' perceptions and experiences comprehensively. For instance, the non-user-friendly interfaces, with a lack of comprehensive life cycle assessment capabilities and seamless integration with existing workflows, hinder widespread adoption. Additionally, many architects perceive sustainability-focused tasks, such as BPA, as beyond their professional responsibilities, further limiting the tools' application (Yang et al., 2023). Notably, there is insufficient exploration of how digital tools can better support the early stages of conceptual design, where the greatest opportunities for sustainability impact exist. Therefore, an indepth understanding of architects' needs and the role of digital environments in facilitating sustainable outcomes is essential.

Potential of hybrid digital tools

The fragmented nature of the architecture industry, coupled with varying technological adoption rates, presents challenges in creating a standardized approach to digital education. Studies suggest that hybrid digital toolkits, which integrate diverse functionalities like shaping, animation, and generative modeling, can be helpful in overcoming these limitations. According to Yu et al., (2021, 2022) and Zboinska (2019), allowing seamless transitions between different interaction modes—human-dominated, tool-dominated, and balanced—these tools can enhance design cognition and creativity. While theoretical models for hybrid digital tools exist, empirical studies on their application in architectural education remain sparse. This study addresses this gap by exploring the practical implications of integrating hybrid digital tools into architectural curricula.

Methodology

A multi-method qualitative approach was employed, combining semi-structured interviews and surveys to gather in-depth insights, as shown in Fig. 1. Data were collected from 50 architecture students and 23 experienced academic architects and researchers. Qualitative research provides in-depth and rich information to study the effects of using hybrid digital tools on the design outputs of architecture students. This approach ensures that the academic position is understood in an integrated manner to determine the impact of new interventions and tools on the efficiency of designed outputs, allowing for an improved initial understanding of the phenomena being studied and their different effects (Bailey et al., 2020; Maxwell, 2009).

Given the exploratory nature of this study and the potential for diverse meanings to different people, semi-structured interviews were deemed appropriate. This format allows for flexibility while ensuring that essential questions relevant to the study goal are asked (Crone & Lozano-Sufrategui, 2018; Kallio et al., 2016) The study also included a survey to determine the current situation in architectural schools and the level of overlap between application and digital design tools during architectural product development. Next, using a semi-structured interview method, this research surveyed experienced academic architects and researchers at accredited program to discuss the implications of using a hybrid toolkit for the architectural design process in integrated design courses and how the current digital architecture practice has been evaluated.

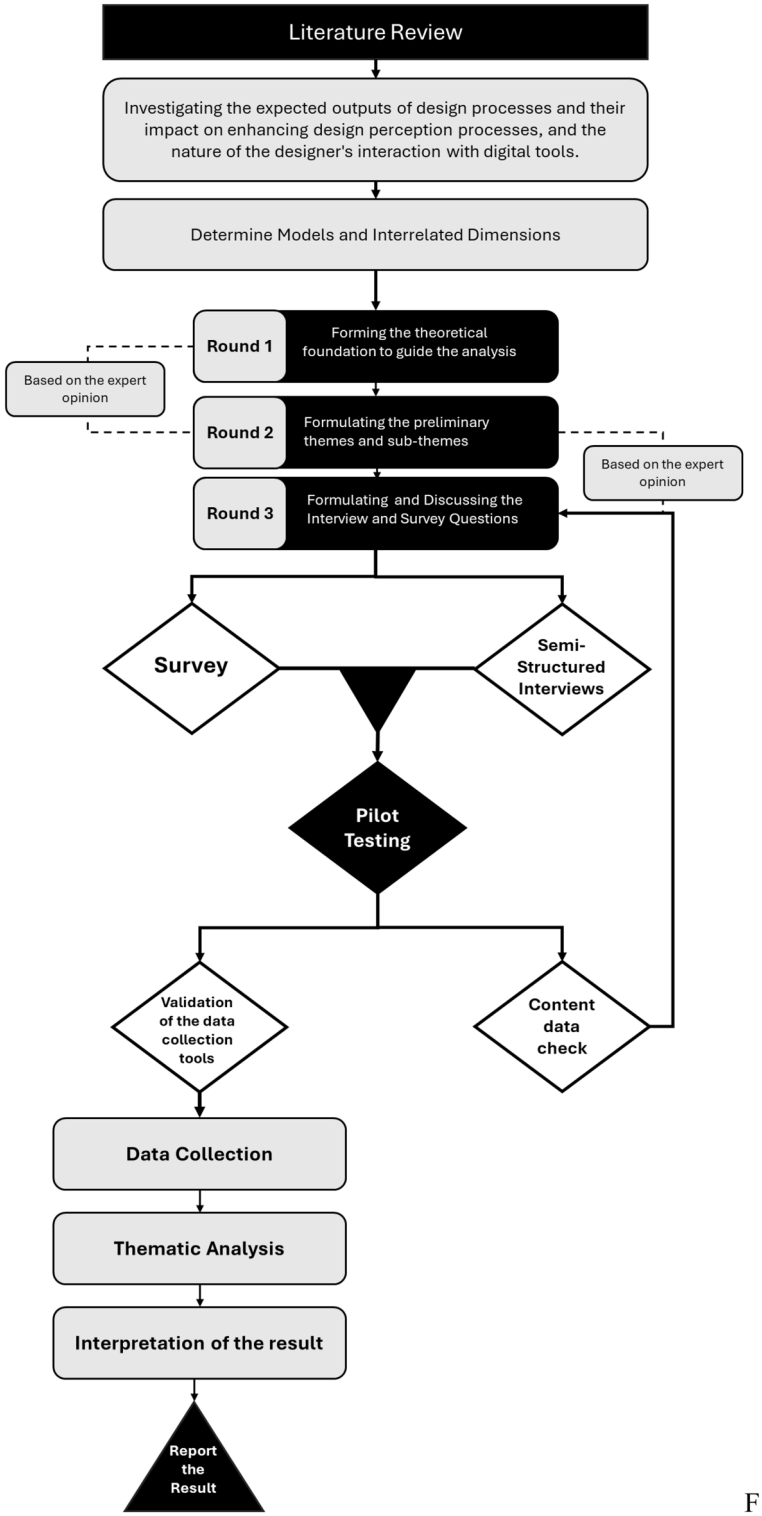

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Fig. 1 Methodology flow chart (Author)

Survey questions and semi-structured interviews guide

An interview guide was followed to develop interview and survey questions based on the current literature (Kallio et al., 2016). Possible follow-up questions and prompts were included in each interview to keep the discussion focused on the research goal (Crone & Lozano-Sufrategui, 2018; Kallio et al., 2016) The framework for developing the semi-structured interview guide and survey is described in Fig. 2. The interview and survey questions developed for this study aim to explore the impact of digital tools on various aspects of the design process. These questions are organized around four interrelated dimensions, each grounded in one of three theoretical models. The first model, Brian Lawson's design process model, identifies five main stages of design: understanding the problem, generating ideas, developing and evaluating these ideas, and finally, representing the design. These stages are interlinked, illustrating how problems and solutions evolve throughout the process. The second model, Oxman's complex integrated digital design model, focuses on the designer's interaction with five distinct 3D modeling methods, highlighting the intricate relationship between digital tools and the design process. The third model, Yu et al.'s design practice model for computational design, explores the interactions between the design environment, design technology, design cognition, and design creativity, providing a comprehensive foundation for understanding how digital tools influence design outputs. Together, these models form the basis for analyzing the impact of using multiple digital tools across various facets of design.

The main focus areas of the study address how digital tools influence different stages and outcomes of the design process. The first area, coping with internal design tools, examines the skillset and the ability of the designers to select and effectively use the most appropriate digital tools for specific design tasks. The second area investigates the impact on the outputs of the design process, exploring how digital tools affect the five stages outlined by Lawson—from problem understanding and idea generation to evaluation and representation of the final design. The third area delves into the impact on design cognition or the way of thinking. This includes critical thinking, problem-solving, the ability to create and assess design alternatives, and how digital tools support the adaptation and improvement of designs. It also covers the visualization of the final

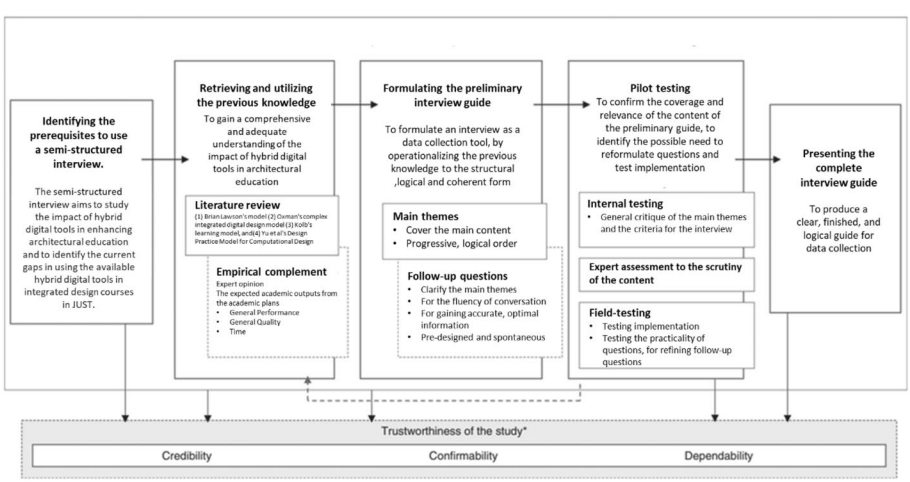

Fig. 2 Framework for developing the semi-structured interview guide (Kallio et al., 2016): Edited by Author)

design and its accuracy in terms of spatial, environmental, and construction aspects. The fourth area evaluates the impact on design quality, focusing on how digital tools enhance the quality of both early-stage proposals and final design outcomes. Lastly, the study explores the impact on design creativity, assessing how digital tools foster originality, novelty, resolution, elaboration, and synthesis in design thinking.

By exploring these areas, the study seeks to understand the broader impact of digital tools on design processes and outcomes. The results will offer insights into how digital tools influence not only the technical execution of design but also cognitive and creative aspects, ultimately guiding designers in optimizing their workflow, enhancing creativity, and improving the overall quality of their designs. This comprehensive analysis, grounded in established design models, aims to contribute to a deeper understanding of the role digital tools play in shaping modern design practices.

Survey

Fifty out of sixty one students registered in an integrated design course were accepted to participate in the survey. The survey was distributed to students during the final and pre-final submission period. The collected data were analyzed using SPSS in combination with Microsoft Excel.

Semi-structured interview

Semi-structured interviews are a popular qualitative research technique that can help guarantee that important questions that are directly relevant to the study goal are asked while allowing some flexibility to provide more in-depth results (DeJonckheere & Vaughn, 2019). The semi-structured interview format, as compared to other qualitative methodologies, maintains a sufficient focus on relevance to the research issues while also being responsive to individual interviewees (Henderson & Marcia, 2000).

Participant selection

Semi-structured interviews were conducted with academics and researchers with experience in teaching integrated design courses 15 academic architects and eight research architects in Jordan were accepted to participate. Participants were chosen based on their experience in integrated design courses and familiarity with hybrid digital tools. This ensured a diverse range of perspectives on the study's central themes. All appointed architects have over seven years of academic experience teaching architectural design courses at all levels. Academic architects and prospective researchers were initially selected based on their academic experience and academic and research specialization. The interviews were conducted in the field over one month. The duration of each interview was roughly an hour, and the sessions were taped for further examination.

Data analysis

Thematic analysis was employed to analyze the interview data. Every interview was recorded and then transcribed. All transcripts were translated into English and back-translated to guarantee accuracy. According to Clarke and Braun's (2013) guidelines, the qualitative data required six steps, as shown in Table 1. The transcribed data were read and reread, and initial thoughts were recorded.

Table 1 Six-step thematic analysis procedure (Clarke and Braun, 2013)

Phases	Description of analysis process
Familiarize with the data	Manually transcribing the data Read and re-read the data from the sources
Generate initial codes	Code interesting patterns of the data systematically for one Architect Deductively code patterns based on pre-determined theory Collate data relevant to each code
Search for themes and subthemes	Collate codes into themes Gather data relevant to each theme
Review the developed themes	Check the themes for their relevance to the coded extracts Check the themes for their relevance to the entire data set Review data for additional themes Generate a thematic map
Define themes	Refine the themes Generate definitions for the themes
Write up	Select vivid and compelling excerpts Review of the selected excerpts relate to the research question, objectives, and literature review Repeat Phases 1 to 6 for the 15 academic architects and 8 research architects till complete Produce a report

- Creating initial codes: The transcripts were coded manually.
- Finding themes: The codes were put together to create themes based on the research questions and suggestions from the expert group.
- Examining themes: Each extracted topic was examined to see if it overlapped with another theme.
- Themes' definition and naming: Each theme was precisely described and given a name.
- Making the report: Each discussion was held in the context of a sub-theme.
- The steps indicated by Clarke and Braun (2013) were followed in this study to begin the data analysis process.

Main themes for encoding data

By applying Brian Lawson's model, Oxman's complex integrated digital design model, and the design practice model, five comprehensive themes were identified to classify the effects of using more than one digital tool on the expected outputs of different design processes. These themes were further divided into sub-themes based on the impact on design cognition, design quality, and design creativity.

Table 2 shows the detailed division of factors under the five main themes: coping with the internal design tool, impact on the outputs of the design process, impact on design cognition—way of thinking-, impact on design quality, and impact on design creativity.

The factors related to the use of interior design tools refer to the mechanism of the designer's interaction with the digital design tools and the skills required to use them effectively, including proficiency in tool usage, the tool's capabilities, and the ability to choose the appropriate tool to perform the tasks. Additionally, factors related to impacts on the way of design cognition -way of thinking- through the use of digital

Table 2 Detailed division of factors (author)

Main themes	Detailed division of factors
Coping with the internal design tool	Digital skillsets, capabilities of digital tools, and selection of the appropriate tool
Impact on the outputs of the design process	Understanding a design problem, generating ideas, developing the idea, evaluating the ideas, and representing the design
Impact on the way of design cognition -way of thinking	Critical thinking, problem-solving activity, ability to create design alternatives, ability to assess the appropriate design, design adaption process, ability to improve the design product, improve the integrity of the final design, visualization of the final design, experience, and the sense of the element through the final design, and accuracy of spatial design in the final design, and visualize the environment, construction, and planning decisions in the final design
Impacts on design quality	Quality of the proposed solutions in the early stages of design and the quality of the final design in the final stages of design
Impacts on design creativity	Novel ideas (Original and surprise), Resolutions (Valuable, logical, useful, understandable), elaboration, and synthesis

design tools include their influence on critical thinking, problem-solving, creating design alternatives, assessing the appropriate design, adapting design, improving the design product, improving design integrity, visualizing final design, ensuring spatial accuracy, and visualizing the environment, construction, and planning decisions in the final design. Furthermore, factors related to the impacts on the outputs through the use of digital design tools include their effect on understanding a design problem, generating ideas, developing ideas, evaluating ideas, and representing designs.

In addition, factors related to impacts on design quality through the use of digital design tools include their effect on the quality of the proposed solutions in the early and final stages of design. The factors related to design creativity when using digital design tools and its impact on the ability to generate novel ideas and make valuable, logical, useful, and understandable design resolutions with appropriate elaboration and synthesis.To classify the architects' opinions of the effect of using multiple digital design tools in the integrated design course, four viewpoints were proposed- negative, positive, current, and future as shown in Fig. 3. These categories provide additional insight into the positive and negative experiences of academics and researchers with the subthemes that fall under each of the five major themes. They also define opinions on using current digital tools, associated difficulties, and potential future opportunities. Negative and positive experiences were coded according to participants' general views on the effects of using hybrid digital tools on the architectural outputs of students. Current refers to interview content that relates to the current use of digital tools to extract current gaps. The future refers to the participants' opinions and expectations about the planned or expected opportunities regarding the effects of using hybrid digital tools on the architectural outputs of students. These four categories, along with the five themes, were used to code the collected data and generate the subsequent results.

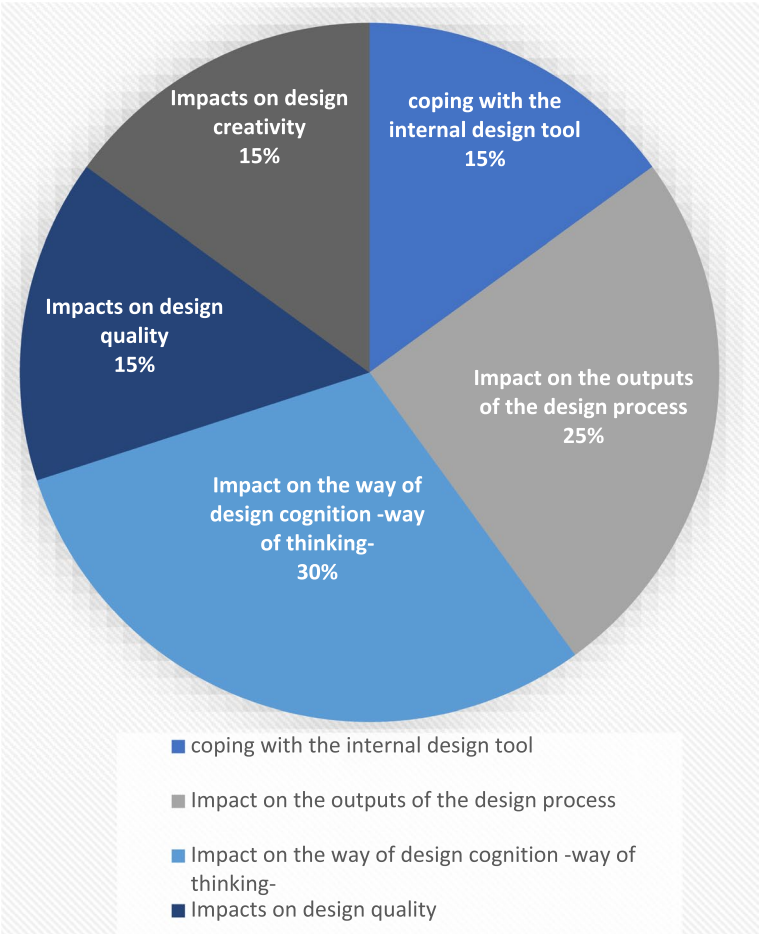

Fig. 3 Coding proportion shows the overall division of factors in terms of the five main themes (Author)

Result and discussion

Distribution of factors affecting design processes with hybrid digital toolkits

This section describes the distribution of factors associated with the five main themes that classify the effects of using multiple digital tools on the design processes. Figure 3 shows the general distribution (%) of factors discussed. The distribution confirms that architects discussed the effects on design cognition -way of thinking- (30%), followed by the impact on design process output (25%), the mechanism of dealing and interaction with interior design tool (15%), and finally effects on design quality (15%) and design creativity (15%). This distribution highlights the significant impact of using digital tools on design cognition, particularly critical thinking and problem-solving activity, and the creation of design alternatives. These aspects are central to the design process, demonstrating how design techniques, tools, and technology are used to showcase design creation processes.

Figure 4 shows the detailed distribution of factors for the five main themes. The results show that "digital tool capabilities" and "final design quality at the final design stages" have the most significant coding ratio (9%). Followed by "represent the design," "develop the idea" (7%) "Generate ideas," "Quality of solutions proposed in the early

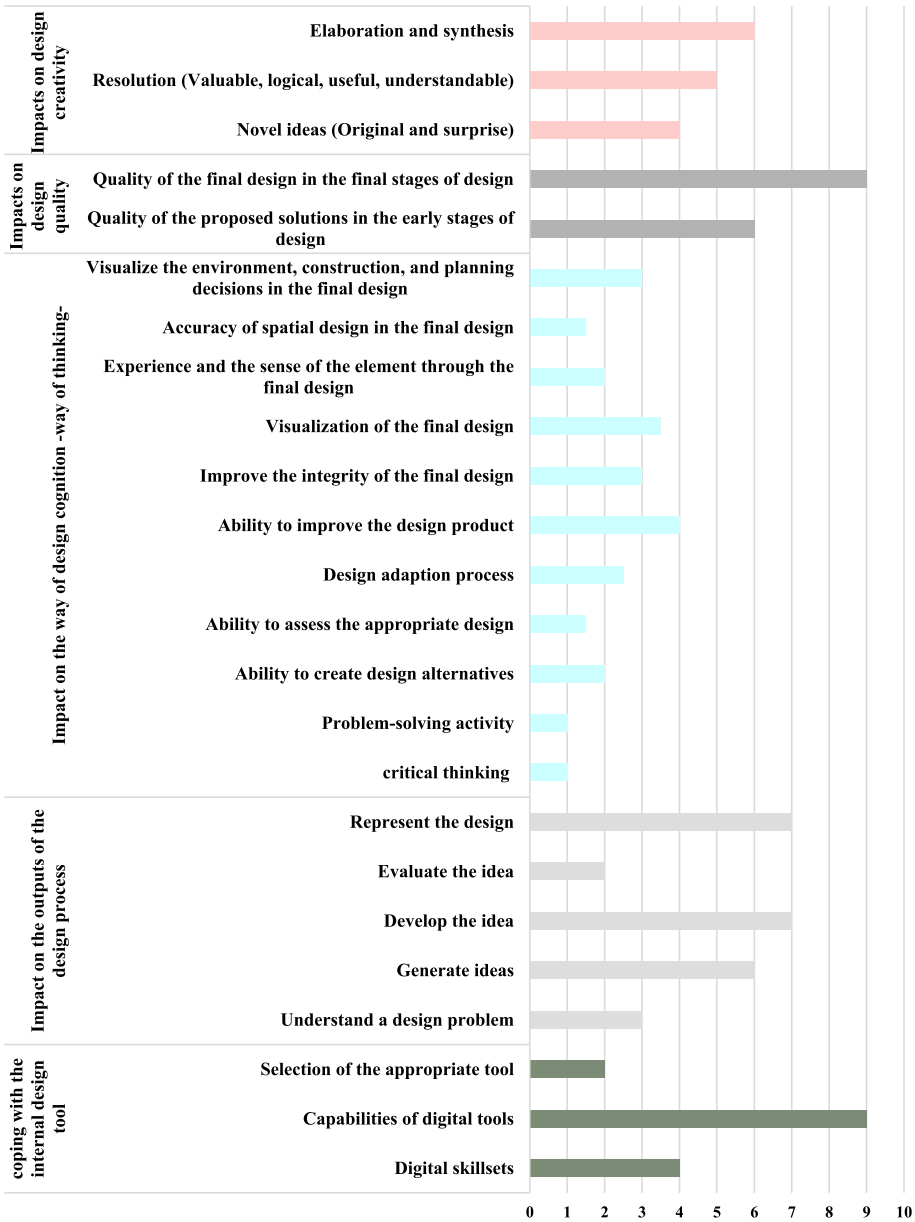

Fig. 4 Coding proportion shows the detailed distribution of factors in the five main themes. (Author)

stages of design" and "Detail and synthesis" (6%). These factors are often linked to interior design tools and their direct effects on design quality and design creativity.

These findings imply that, despite the potential dominance of digital design techniques in influencing the integration of design processes, they are not effectively utilized to support architects' design processes. However, this may also be a result of the interview questions' tendency to stress perceptions of digital design processes in a manner that progresses from the most general to the most private. Among the factors associated with design cognition and thinking, the "ability to improve product design" represents a key element in the design process. This factor highlights the interaction between digital tools and the designer and the role of both in controlling the design process. Participants

emphasized the importance of this factor, noting that it reflects the integration between mental processes, concrete movements, and resulting design elements.

Another significant factor identified was "Digital skill sets," which was noted as highly influential (4%) during the design process. This factor underscores the importance of designers' proficiency in using digital tools effectively. Additionally, "Capabilities of digital tools." Were highlighted as crucial in detrminig the success of the design process, further emphasizing the need for tools that support and enhance the design process.

While some academics discussed the limitations of digital tools and their potential impact on the quality, performance, and integrity of the design process, others believed that these limitations could be mitigated if designers were able to balance their role with that of digital tools. This suggest that the effectiveness of digital tools in the design process depends on how well designers can integrate them into their workflow.

Within the factors related to design cognition, several key areas were discussed by participants. These included the "ability to improve product design," "improve the integration of final design", "conceptualize final design," and "conceptualize environmental, construction, and planning decisions in the final design." these factors were highlighted as central to the design process representing 4%, 3.5%, and 3% of the discussion respectively. This indicates the importance of these factors in influencing the outcomes of the design process.

With the increasing use of digital tools for architectural design, it is possible to maximize their utilization in fostering both design creativity or innovation and evidence-based design, which are essential to future design practices. This can be achieved by redefining the relationship between designer and modern digital tools and exploring the possibilities for their overlap in design processes.

Figure 5 shows a word cloud generated from interview data. The word cloud highlights that the term "Integrated" is the most frequently appearing term, followed by "problem-solving," "Defines the design digital tool," "capabilities of digital tools," "Human perception," "Dynamic," "Response," and "connection". This observation suggests that the between tools and designers that support design thinking is more influential in architectural design outcomes than the digital tools themselves. Participants emphasized the

Fig. 5 Word cloud (Author)

importance of tools complementing mental processes with concrete results and the cognitive process of design elements. " they noted that relying solely on a specific digital tool throughout the design process stages of design eliminates the designer's response to the dynamic of the design process and restrict creativity "; "Human creativity cannot be achieved through a single digital tool, the limited capabilities of the tool will limit human creativity if relied upon rigidly. "Supporting design thinking and creativity is one of the most frequently used phrases when discussing academics, even though it is not the most common component in terms of coding percentages of participants' texts. These findings align with previous research emphasizing the crucial role of the designer-tool interaction in fostering design thinking and creativity.

The perceptions of the effect of using a hybrid digital toolkit for the architectural design process in integrated design courses

To establish a reference context for discussion of the study's findings to conclude the implications of using a hybrid toolkit for the architectural design process in integrated design courses, how the current digital architecture practice has been considered for evaluation, and to find the gaps. The current method of design thinking using digital tools will be presented briefly.

Current perception of the digital design process in architectural schools

Currently, the architectural design process relies on limited digital tools. Designers often begin the early stages of the design process, traditionally focusing more on the final product without considering the thinking process that led to it. This approach restricts the designer's ability to make appropriate adjustments and compromises the interaction between the end user and the designed elements. Participants indicated that reliance on digital tools in recent years has caused a decline in the quality of designs in the early stages and the final product, and thus, a decline in creativity and design thinking.

Most participants indicated that the current method of using digital tools reduces students' and designers' ability to understand and identify architectural design problems, affecting their problem-solving abilities. Participants also indicated that sometimes the conceptual phase of design is sometimes overlooked because it is not supported by the tools used. This has led to a decline in students' spatial and structural thinking skills, their capacities to respond to design problems, and their overall design thinking abilities.

Participants also noted that relying on a specific tool during the design process led to inflexibility in creating proposals and design alternatives in the early stages. Design development became slow, and proposals were often rejected due to a lack of comprehensive control over the design process. Digital tools with limited capabilities began to dominate the designer's decisions and responses to design problems. Consequently, most participants observed a decline in the analytical quality of solutions proposed in the early stages. They unanimously agreed that designers have become less responsive to users' human experiences, resulting in a loss of sensitivity in designed spaces. This shift has also led to decrease in considering complementary design elements and evaluating them effectively. This has led to the emergence of non-human-oriented designs that fail to address human and psychological needs, often producing purely functional designs. It was emphasized that the role of the architectural designer is to respond to and interact

with the materiality of the building or the designed component, the design is not the final model but encompasses decisions related to space and materials. Design is not only about shapes, models, and images; it must also be sensitive to the function for which it is designed.

Current versus hybrid digital toolkits in architectural design

After providing a brief overview of the current perception of the digital design process in architectural schools that are characterized by the use of a limited number of digital tools, it can now be compared with the perception of the impact of using a set of hybrid digital toolkit on design process in architectural school. This comparison focuses on two main aspects: the gaps in the current situation and the fundamental features of the design process. Specifically, this research examines the nature of interactions between the designer and digital tools and the extent to which these interactions influence design thinking.

Nature of interactions with digital tools

To support design thinking and creativity in digital design, tools should provide three forms of interactions: a human-dominated, tool-dominated, and a balanced collaboration between humans and tools. One participant indicated that a digital tool or any other tool must be responsive to the human mind, linking mental behaviors and design decisions in a balanced way, not making decisions that are merely responsive to the tool. Currently, the digital design process achieves one or at most two forms of interaction with digital tools, with limited and ineffective transitions. This inflexibility of the interactions between the designer and the digital tool restricts comprehensive solutions to design problems and poses a danger to the designer's creativity and design thinking. Surveys indicate that using hybrid digital toolset supports all three forms of interaction,creating a diverse interactive environment that positively effect design thinking. It allows designers to achieve a sequential balance between intuitive, manual, and technical processes to support activities that are difficult to perform manually while maintaining the designer's central role. In contrast, current digital tools often replace traditional and manual design methods and thus lose the direct physical link between manual intuitive processes and the brain, as many researchers have warned (Cross, 2021; Soliman et al., 2019; Xiang et al., 2020a; Yu et al., 2022).

Some participants highlighted that the efficacy of using the hybrid toolkit depends largely on the designer's proficiency with digital tools, enabling them to make a seamless transition between the tools and different forms of interaction. This familiarity empowers designers to use the tool's capabilities effectively, enhancing their awareness of leveraging digital tools and their potential for integration throughout the design process. Conversely, current limitations in tools and their imposed interaction constraints significantly restrict such interactions during the design process.

Impacts on the design thinking process

The findings indicate that hybrid digital tools significantly enhance design cognition by supporting critical thinking and problem-solving activities. Participants highlighted the tools' ability to facilitate the iterative exploration of multiple design solutions, a process

often restricted by traditional digital tools. Survey and interview responses revealed that current digital tools are predominantly task-specific, focusing on detailed design stages while neglecting the conceptual phase. This limits students' ability to generate and evaluate creative design alternatives. Creative design using digital design tools should include three types of design thinking activities and stages: divergence, transformation, and convergence, and should be implemented in cycles to enhance and ensure that the design results are meaningful and applicable to produce a final design concept.

Currently, design thinking depends on transformations for a single design solution. In contrast, a hybrid digital toolkit allows for multiple cycles of creating and evaluating solutions using more than one digital design tool. Examining the impact on the five basic design processes—formulation, representation, moving, evaluation, and reflection, reveals that the nature of the interactions achieved by the hybrid toolkit positively supports the outputs. The participants also pointed out its ability to address current gaps in the method of dealing with design thinking by encouraging exploratory solutions and integrating mental intuitive procedures with technical capabilities. This facilitates testing and comparing solutions, supporting innovation and creativity sensitive to design problems.

A hybrid toolkit supports the creation of different alternatives and design proposals, which may expand the space for solving design problems and stimulate innovation and creativity sensitive to designed solutions. Moreover, this is where using a hybrid digital toolkit contributes to changing the role of tools to become real partners for the designer and catalysts for the creative processes in design thinking, unlike the classic role of tools in the current situation. Creativity involves developing existing models and producing innovative solutions to design problems.

Strategies for enhancing digital tool education in architecture

The participants recommended the need to develop general strategies for teaching digital tools to ensure achieving a high level of interaction and control over tools to serve design thinking and design requirements in general. Furthermore, bridging the architectural education gap involves integrating theoretical, applied, and technical knowledge into curricula to enhance creative thinking.

The study suggests integrating hybrid digital tools into architectural curricula. Specific strategies include Introducing hybrid tools in conceptual design modules to foster creativity, providing training programs to enhance digital proficiency among students and educators, and encouraging iterative feedback mechanisms that leverage the strengths of hybrid tools.Course content focused on digital applications should be carefully designed to align with different stages of the design process. Focusing on improving designers' control over tools to increase design efficiency and support design thinking processes. However, contrary to some studies, integrating digital tool education within design courses may unintentionally restrict thinking and problem-solving methods if not implemented thoughtfully.

It is recommended that the content of the courses involved in teaching digital applications should include hybrid tools that take into account the different stages of design. Improving designer control over tools increases design efficiency and serves design thinking while solving a design problem. Contrary to some studies, integrating digital

tool education within design courses may unintentionally restrict thinking and problem-solving methods.

To address the limitations of existing digital tools, Educators should integrate hybrid digital tools into architectural curricula, particularly during the concept stage of design, where creative thinking is most critical. Effective integration of the hybrid tools can be achieved by following several strategies, such as providing specialized training programs to improve digital proficiency among students and educators, fostering iterative feedback mechanisms, and designing course content to incorporate tools that align with different stages of the design process.

On the other hand, practitioners are encouraged to adopt hybrid tools that support design thinking and problem-solving processes by facilitating interaction, experimentation and smoother transitions between traditional and digital methods. Additionally, incorporating advanced technologies during the early phases of design can further enhance interaction and creativity. These approaches aim to empower designers with greater control over tools, improve design efficiency, foster innovative outcomes, and advance digital tool education and practice in architecture.

Future research should focus on increasing students' interaction with technical and digital design tools. Studies should also explore the effect of using some digital tools in site analysis and the initial data and the possibilities of integrating interactive digital tools that have the potential to allow design experimentation and interaction with the design creation process, such as extended reality techniques in the early stages.

Conclusion

This study aimed to examine the impact of hybrid digital tools on architectural education and to identify the current gaps in the usage of digital tools in integrated design courses at NAAB accredited program through thematic analysis of 23 semi-structured interviews with academic architects and researchers. The findings demonstrated that using digital tools significantly impacts the processes naturally controlled by the designer, as well as the overall perception of design and general thinking, including critical thinking, problem-solving, and the capacity to generate design alternatives.

Over-reliance on digital tools in recent years has led to a decline in the quality of designs in the initial stages and the final product. This decline in creativity and design thinking is attributed to the insufficiently interactive use of digital tools, which affects the integration of design processes. Reframing the relationship between the designer and contemporary digital tools and their potential overlap in design processes can maximize their use in encouraging both design creativity or innovation and evidence-based design. These results indicate that the factors related to the nature of the interaction between the tools and the designer support design thinking more than the digital tools themselves. Thus, it was emphasized that human creativity could not be achieved through a single digital tool, as its limited capabilities will constrain human creativity if relied upon exclusively. The study highlighted that the weaknesses and limitations of the current interaction methods restrict the ability to reach comprehensive solutions to the design problem. Also, the inflexibility of the interactions between the designer and the digital tools threatens creativity and design thinking. Therefore, using a hybrid digital toolkit supports the three forms of interaction and

different design processes. This enables the designers to achieve a sequential balance between intuitive, manual, and technical processes, supporting activities that are challenging to complete manually while maintaining the necessary direct physical link between manual and intuitive processes and the brain. This is unlike the current situation where digital tools often replace traditional and manual design tools, leading to the loss of the direct physical link between manual intuitive processes and the brain, as emphasized by many researchers.

Finally, the results indicated that the use of the hybrid digital toolkits allows for the integrated inclusion of the three stages of design thinking in an integrated manner, through the possibility of creating and changing solutions multiple times using various digital tools and evaluating results side by side. Further studies should focus on improving the way hybrid digital tools are used among undergraduates, particularly in design development, instead of just conceptual and schematic design. Additionally, it is essential to assess the education process to ensure it aligns with the field of practice, is aware of digital tools applications, and explores ways to increase students' interaction with hybrid digital tools. This study underscores the transformative potential of hybrid digital tools in architectural education. By bridging the gap between traditional and digital methods, these tools facilitate a holistic approach to design thinking, enhancing creativity and cognitive processes. The findings advocate for a significant shift in architectural education, emphasizing the integration of hybrid tools to address existing limitations and foster innovation.

The findings have significant implications for integrating hybrid digital tools for key stakeholders, including educators, curriculum designers, and policymakers. For educators, the findings highlight the need for specialized training and resources to effectively integrate hybrid digital tools into design curricula, thereby enhancing students' creativity and cognitive abilities.

Curriculum designers are encouraged to develop modular courses that integrate hybrid tools across various stages of the design process to promote iterative and explorative processes while aligning with desired educational outcomes. For policymakers, the study emphasizes the importance of developing frameworks to facilitate the implementation of innovative technologies in architectural education while simultaneously prioritizing creativity and innovation.

Future research should explore the long-term impact of hybrid tools on students' creativity, cognitive process, and learning outcomes. As well as evaluating the feasibility of implementing these strategies across diverse educational settings.

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Declarations

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